

Public benchmark pilot: results

Evaluation 3: USC xArm public pilot

Completed 54 paired timestamps across 30 evaluation attempts (six tasks), with six separate successful reference demonstrations. Native scores only; no calibration.

Method	Terminal	tau-a	Success AUROC	Success VOC
PSI DT	0.139	0.500	-0.005	
PSI S1	0.472	0.750	0.091	
PSI S3	0.500	0.792	0.157	
PSI S5	0.472	0.750	0.148	
Robometer progress	0.722	1.000	0.962	
Robometer success	0.722	1.000	0.864	
DINOv2	0.389	0.792	0.516	
Encoded time	0.000	0.500	1.000	
Constant	0.000	0.500	undefined	

Task-bootstrap intervals and individual tasks are in metrics.json / task_metrics.csv. With only six tasks and one evaluated successful attempt per task, uncertainty is large; this pilot is not a full published-table comparison.

Independent goals share task/camera/setup, but target/distractor layout and arm poses differ. Public clips are resampled to 64 frames; original durations are unavailable. Five success prefixes use frame indices 4,15,31,47,63; initial warmup and intermediate prefix rounding differ from the upstream example. Robometer has language while PSI/DINO have visual goals. No complete training-membership audit is possible from configuration names alone.

Terminal quality-triplet Kendall tau-a reproduces the pinned upstream function exactly on actual scores and tied controls. Secondary pair accuracy gives ties half credit; upstream pair-accuracy code treats ties asymmetrically by input order, so we do not reproduce that artifact. Encoded time is constant at all terminals and therefore has chance outcome AUROC and zero quality tau; its successful-trajectory VOC is tautologically one.

Protocol: ccwm/rwm/evals/progress_tracking/EVAL3_PROTOCOL.md. Raw source pins/checksums are in ../download_receipt.json.

Frozen public benchmark protocol

Evaluation 3: public Robometer benchmark adaptation

2026-09-20: Atlas explicitly requested a separate agent execute this concurrently with synthetic ABC evaluation 2. This supersedes the previous strictly serial stage ordering for evaluation 3.

Frozen first run: USC xArm

Code: ``ccwm/rwm/evals/progress_tracking/eval3/``. Large artifacts: ``ccn2b/dataset/brd/rwm_progress_tracking/eval3_public/xarm_v1/``. The full six-scene public suite is not complete.

Public source ``aliangdw/usc_xarm_policy_ranking``, pinned revision ``42468f8f464120a110f54ed6307172b215ce6724``. It is in the upstream ``rbm-1m-ood`` dataset group. The parquet contains 36 trajectories across six tasks, each with two failure, two suboptimal and two successful labels. Two additional video files have no parquet row and are not evaluated. Released quality annotations are used unchanged. Raw public USC Franka and UTD top data were also downloaded for access/goal auditing; they are not part of the first scored pilot.

For each task, select one successful trajectory by a fixed seeded hash of its source ID as the reference pool. Exclude all six reference episodes from every metric. Every remaining attempt gets that task's reference final frame; no evaluated episode supplies its own goal. Evaluate 30 attempts: 12 failure, 12 suboptimal, six successful. No development fitting, thresholds, fine-tuning, or score-direction selection on these labels.

****Goal limitation:**** visual inspection of all 36 endpoints confirms a shared camera/table/task setup, but distractor and target object positions, arm poses, and towel folding configurations differ. These are independent task-reference goals, not exact matched-scene counterfactual goals. This measures transfer under that limitation. A labeled successful demonstration is used only to supply a visual goal. Robometer instead receives the released task-language instruction, so conditioning is unequal by design and disclosed.

Inference

PSI checkpoint ``psi_0.5_r_abc_v2`` step 820000; restore and verify all embedded config fields. Reuse audited serialization and tokenizer from ABC pilot: resize RGB to 256x336; same seeded 128 patches per image; recent one/three/five frames at consecutive released indices; future ordinary-RGB target at offset five slots; separate goal-role DT. Higher completion scores are negative expected DT and negative goal NLL. DT is not converted to seconds.

Robometer-4B and DINOv2 model revisions remain those in stage 1. Robometer uses the official model/collator and up to eight uniformly selected causal-prefix frames, with the instruction only. This pilot floors intermediate frame indices; upstream rounds. Terminal samples are identical (0,9,...,63), but intermediate VOC inputs can differ by one released frame. DINOv2 uses native current/independent-reference CLS cosine similarity. All observations are causal; reference frames come from disjoint episodes.

The release uniformly subsamples videos to at most 64 frames, encoded at 10 fps. No original robot-duration/index mapping is present in the parquet. Save source IDs, paths, released frame indices, encoding FPS and source hashes; do not imply that encoding timestamps represent original physical time.

Score the terminal frame of all 30 attempts. Also score successful-attempt prefixes ending at

Frozen public benchmark protocol

released indices 4,15,31,47,63. The first index is 4 to provide five observation frames; the upstream example VOC grid starts at 0. This warmup difference prevents claims of exact numerical reproduction. There are 54 unique episode/timestamp samples.

Metrics and comparability

Primary ranking: use pinned upstream quality-triplet mean Kendall tau-a within each task, then task-macro average, on terminal scores. This differs from applying one global Kendall coefficient with same-class ties. Higher label order is failure < suboptimal < successful. Validate the local result directly against the upstream function on actual scores and tie controls. Report task-level values and task-bootstrap intervals, with six tasks explicitly noted.

Secondary: terminal success-vs-other AUROC per task and task-macro; cross-quality pair accuracy with score ties contributing one-half; VOC Pearson correlation between native score and released frame index on the six successful trajectories. Constant/time controls are reported; normalized encoded time is a privileged ordering diagnostic because all public clips have the same 64-frame cap, not a usable original-duration predictor. Per-trajectory score normalization is not used.

This is a public-data native-score adaptation/pilot, not reproduction of the full published table: only one of six scenes, six held-out goal demonstrations, different first VOC prefix, visual versus language conditioning, unknown complete pretraining membership, and independent-goal layout variation. No claim of zero training overlap can be established solely from checkpoint dataset names. Audit the embedded training mixture for explicit source matches and record the limits separately.

Sources: <https://github.com/robometer/robometer> at `352d160389daa964788delec933d1925f3a6de4f`; <https://arxiv.org/abs/2603.02115> (v2); public dataset linked above. Dataset download receipts include SHA256 and exact revision, with size/LFS hash checks. Code and artifact hashes are recorded on completion.

Completed first-run results

All 54 samples scored successfully with PSI DT/S1/S3/S5, Robometer progress/success heads and DINov2. Terminal task-macro tau-a / success AUROC: DT 0.139 / 0.500; S1 0.472 / 0.750; S3 0.500 / 0.792; S5 0.472 / 0.750; Robometer progress 0.722 / 1.000; DINov2 0.389 / 0.792. Success-prefix VOC: DT -0.005, S3 0.157, Robometer 0.962, DINov2 0.516. These results show the strong matched-goal ABC DT finding does not transfer directly to independent-layout public references. They do not isolate whether the change is caused by domain shift, reference mismatch, or temporal resampling. Only six evaluated successful trajectories contribute to AUROC/VOC; see task-bootstrap intervals in metrics.json.

Validation: 54/54 unique paired outputs for every model; no duplicate source-video hashes; six goals disjoint from all 30 evaluated episodes; causal index checks; restored checkpoint configuration parity; direct exact upstream tau parity on actual scores and order/reversal/tie controls. No model or fusion calibration used.

Source access audit: all six upstream OOD scenes are publicly accessible without gating. Remaining five scenes are not scored by this pilot. UTD provides only one successful example per task, so a fully independent reference plus an evaluated success cannot be constructed from that scene alone.